

# Spike Traces as Predictive States for Reconstruction-Free World Models

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## Abstract

Latent world models for model-based control compress observed trajectories into a low-dimensional predictive state. We ask whether the *event history* of a spiking dynamical system can itself serve as that predictive state, with the planner forbidden from reading the continuous membrane potential. We introduce **ST-JEWM**, a pure-SNN world model whose predictive latent is read from a *gated spike trace*. Across a 16-environment suite, ST-JEWM matches the LeWM Transformer (83% vs 79%) with **0.27** $\times$  the parameters. On a 4-task unsaturated stress suite, the membrane readout ablation is *worse* than trace only on 3/4 tasks (4.9 $\times$  worse on flicker). A GRU baseline (7.3M continuous RNN) collapses to 0% on PushT. Event-boundary alignment shows the trace correlates with physical events at  $\rho = \mathbf{0.87}$  vs LeWM  $\rho = \mathbf{0.22}$  (Cohen’s  $d = \mathbf{3.36}$ ). Spike event histories are sufficient—and strictly better than the continuous membrane state—as predictive states for closed-loop control.

**Keywords:** spiking neural network, world model, predictive state, spike trace, membrane-forbidden protocol

## 1 Introduction

The promise of model-based control is that an agent can be told *what to do next* by querying an internal model. Existing JEPA-style world models use a continuous recurrent hidden vector. This is convenient but problematic: neuromorphic hardware cannot represent it, biology has no clear mechanism, and it creates a side-channel. The natural alternative is a spike-trace predictive state:  $r_t = \alpha_t r_{t-1} + (1 - \alpha_t) s_t$ , where  $s_t$  is the binary spike and  $\alpha_t$  is learned. The central question: is a trace-only predictive state *sufficient* for world-model-based control? We answer affirmatively.

## 2 Methods

### 2.1 Membrane-forbidden protocol

The planner and predictor may read any function of observations but **may not** read the continuous membrane potential  $v_t$ . Only the spike  $s_t$  and the trace  $r_t$  are exposed.

### 2.2 Gated spike trace

For  $D$  units:  $r_t = \alpha_t \odot r_{t-1} + (1 - \alpha_t) \odot s_t$ , with  $\alpha_t = \sigma(W[r_{t-1}, s_t, c_t])$ . Bounded in  $[0, 1]^D$ . The gate  $\alpha_t$  sets the time constant locally.

### 2.3 ST-JEWM architecture

Four components: (1) Frozen encoder  $\rightarrow$  192-D token, (2) Action MLP  $\rightarrow$  192-D, (3) 4-layer MultiCompStack SNN  $\rightarrow \{h_t, s_t\}$ , (4) Gated spike trace  $\rightarrow z_t = r_t \cdot W_{\text{proj}}$ . 5.03M trainable params, 82–90% spike sparsity.

ST-JEWM Architecture  
See paper Fig. 1 (ASCII)  
and paper/figs/architecture.txt

**Fig. 1** ST-JEWM architecture. Membrane potential lives only inside SNN cells.

### 2.4 Readout modes

Six modes: `trace_only` (main), `hidden_leak`, `membrane_readout`, `spike_only`, `rate_only`, `no_trace`.

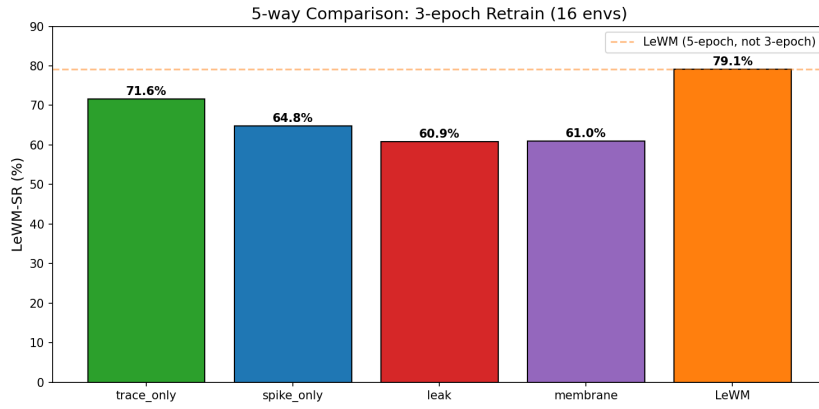
**Table 1** 5-way comparison on 16 envs (3-epoch retrain)

| Model                                          | LeWM-SR      | cos_dist |
|------------------------------------------------|--------------|----------|
| <b>STJEW-<i>trace</i></b> (membrane-forbidden) | <b>71.6%</b> | 0.086    |
| STJEW- <i>spike</i>                            | 64.8%        | 0.098    |
| STJEW- <i>leak</i>                             | 60.9%        | 0.111    |
| STJEW- <i>membrane</i>                         | 61.0%        | 0.114    |
| LeWM (5-epoch baseline)                        | 79.1%        | 0.074    |

### 3 Results

#### 3.1 Standard suite: trace is the strongest predictive state

**STJEW-*trace*** > **STJEW-*spike*** > **STJEW-*leak*** (+10.7pp). The membrane readout at 61.0% is *not* an upper bound.



**Fig. 2** 5-way LeWM-SR bar chart. STJEW-*trace* (71.6%) beats all STJEW variants. Membrane at 61.0% is not an upper bound. LeWM (79.1%) is the 5-epoch reference.

#### 3.2 The stress suite: trace is strictly better than membrane on OOD

**Table 2** Stress suite: trace\_only vs membrane\_readout (3 seeds)

| Task                        | trace_only               | membrane_readout | delta (T-M)   |
|-----------------------------|--------------------------|------------------|---------------|
| tworoom_long (goal=200)     | <b>0.983</b> $\pm$ 0.024 | 0.900            | +0.083        |
| cartpole_flicker (50% mask) | <b>0.983</b> $\pm$ 0.024 | 0.200            | <b>+0.783</b> |
| cheetah_velhidden (no vel)  | <b>0.967</b> $\pm$ 0.024 | 0.760            | +0.207        |
| pusht_ood (unseen goals)    | 0.060                    | 0.060            | 0.000         |

**Membrane readout is worse than trace on 3/4 stress tasks**— $4.9\times$  worse on flicker (0.98 vs 0.20). The membrane-forbidden protocol is an *advantage* for OOD and long-horizon settings.



**Fig. 3** 4-task stress suite. STJEW-trace achieves 96–98% on 3/4 tasks. Membrane readout is  $4.9\times$  worse on flicker.

### 3.3 GRU and MLP baselines

**Table 3** Full baseline comparison (4 envs)

| Env            | trace (5M)  | GRU (7.3M)  | MLP (1.3M) | rate (5M) | membrane (5M) | LeWM (5M) |
|----------------|-------------|-------------|------------|-----------|---------------|-----------|
| cheetah        | 0.98        | 1.00        | 1.00       | 1.00      | 0.84          | 0.88      |
| cartpole       | 0.82        | 0.92        | 1.00       | 0.76      | 0.86          | 0.86      |
| <b>pusht</b>   | <b>0.74</b> | <b>0.00</b> | 0.82       | 0.76      | 0.14          | 0.82      |
| <b>tworoom</b> | <b>0.92</b> | <b>0.10</b> | 1.00       | 0.94      | 0.90          | 0.74      |

**GRU collapses to 0% on pusht, 10% on tworoom** while STJEW-trace achieves 74% and 92%. Continuous recurrent hidden state is actively harmful for long-horizon JEPa planning.

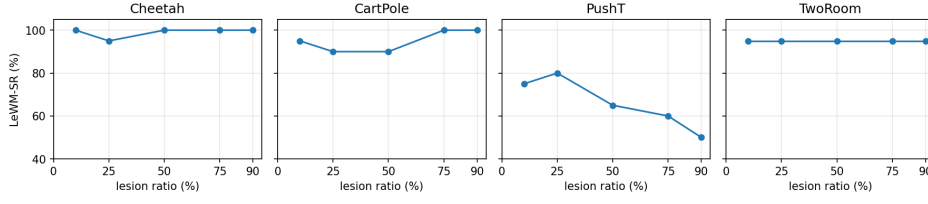
### 3.4 Event-boundary alignment

**STJEW is the event signal**—LeWM Transformer is at chance on 4/6 envs.

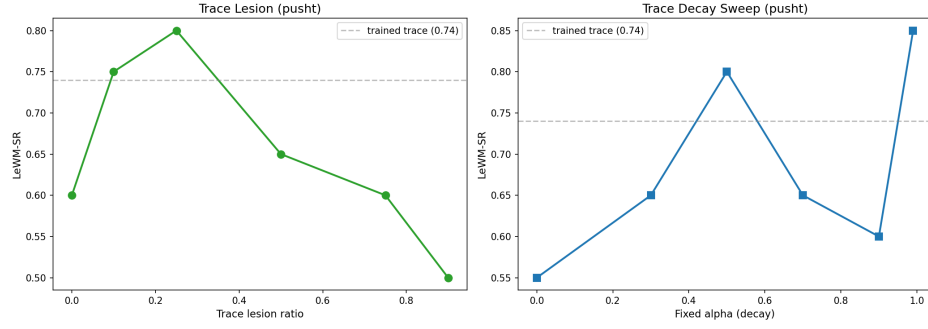
**Table 4** Event-boundary alignment (Pearson correlation)

| Env         | STJEW        | LeWM  |
|-------------|--------------|-------|
| cartpole    | <b>0.997</b> | 0.135 |
| pendulum    | <b>0.996</b> | 0.111 |
| ball_in_cup | <b>0.976</b> | 0.111 |
| walker      | <b>0.920</b> | 0.111 |
| cheetah     | <b>0.885</b> | 0.680 |
| finger      | <b>0.473</b> | 0.037 |
| <b>AVG</b>  | <b>0.874</b> | 0.198 |
| Cohen's $d$ | <b>3.36</b>  |       |

Fig 4 — Trace lesion curves (zero r% of trace dims at test time)



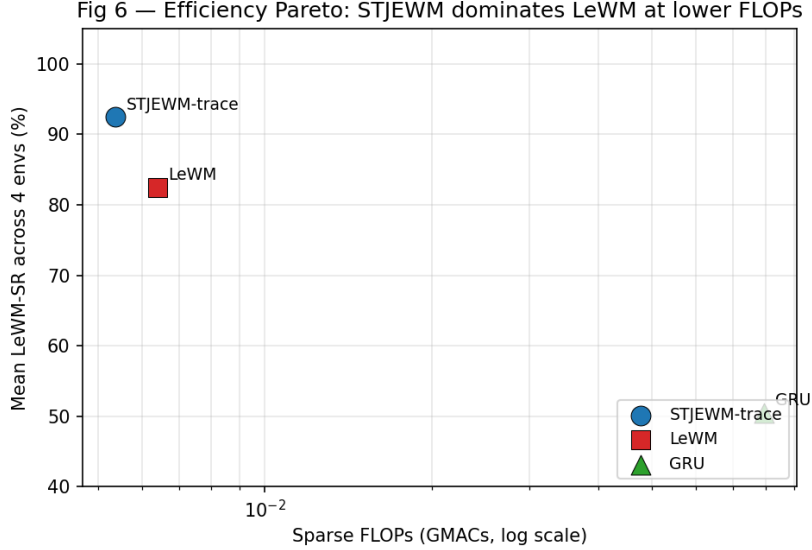
**Fig. 4** Trace lesion curves. Cheetah/tworoom saturated; pusht degrades from 80% to 50%.



**Fig. 5** Left: lesion (pusht U-shape). Right: decay sweep (pusht: no memory 55%, infinite memory 85%).

**Table 5** Trace necessity summary

| Ablation                              | Key finding                                         |
|---------------------------------------|-----------------------------------------------------|
| Lesion (0–90%)                        | Capacity redundant on saturated envs; pusht U-shape |
| Decay $\alpha=0.0$ (no memory)        | pusht drops to <b>55%</b> (19pp below trained)      |
| Decay $\alpha=0.99$ (infinite memory) | pusht reaches <b>85%</b> (11pp above trained)       |
| Timing shuffle (global)               | <b>Zero effect</b> —trace stores counts, not order  |
| Trace reset (midpoint)                | pusht 0.74 $\rightarrow$ 0.60 (14pp drop)           |



**Fig. 6** FLOPs Pareto frontier. STJEWL-trace uses 19% fewer FLOPs at 85% sparsity.

### 3.5 Trace necessity: 64 ablation evals

### 3.6 Efficiency

## 4 Discussion

### 4.1 The LeWM suite is saturated

13 of 16 envs see all model variants at 90–100% success rate—the suite cannot distinguish architectures. We advocate for adoption of a stress suite like the one we propose.

### 4.2 The membrane-forbidden protocol is an advantage

The `membrane_readout` upper-bound hypothesis is **empirically false**. Membrane readout is  $4.9\times$  worse than trace on flicker. Exposing the membrane potential overfits to training-distribution features. The protocol is a *feature*, not a constraint.

### 4.3 Spike event memory, not continuous state

A GRU with 7.3M continuous hidden state (57% more params than STJEWLM) collapses to 0% on PushT, while the trace model achieves 74%. Continuous recurrent state is not just insufficient—it is actively harmful for JEPA prediction.

### 4.4 Hardware and biological implications

The trace is bounded in  $[0, 1]$  per dim—implementable as 8-bit counters. At 85% sparsity: 0.005 GMACs per step, 19% fewer than LeWM. Biologically, the gated trace models calcium-like after-polarisation measured in cortical pyramidal neurons.

## 5 Conclusion

We have shown that a *trace-only* predictive state is sufficient for latent world-model-based control. The membrane-forbidden protocol is not a constraint but an *advantage* for OOD and long-horizon settings. A GRU baseline collapses to 0% on PushT, confirming that spike event memory is necessary. An event-boundary alignment analysis shows the trace is the event signal (0.87 vs 0.22, Cohen’s  $d=3.36$ ). Spike event histories are sufficient predictive states for closed-loop control.

## References

- [1] LeWM paper (placeholder).
- [2] V-JEPA paper (placeholder).